CODE QUEST — SYSTEM MODULE ARCHITECTURE

Placement Preparation, Assessment, and Multiplayer Hiring Ecosystem | Architectural Design Diagram

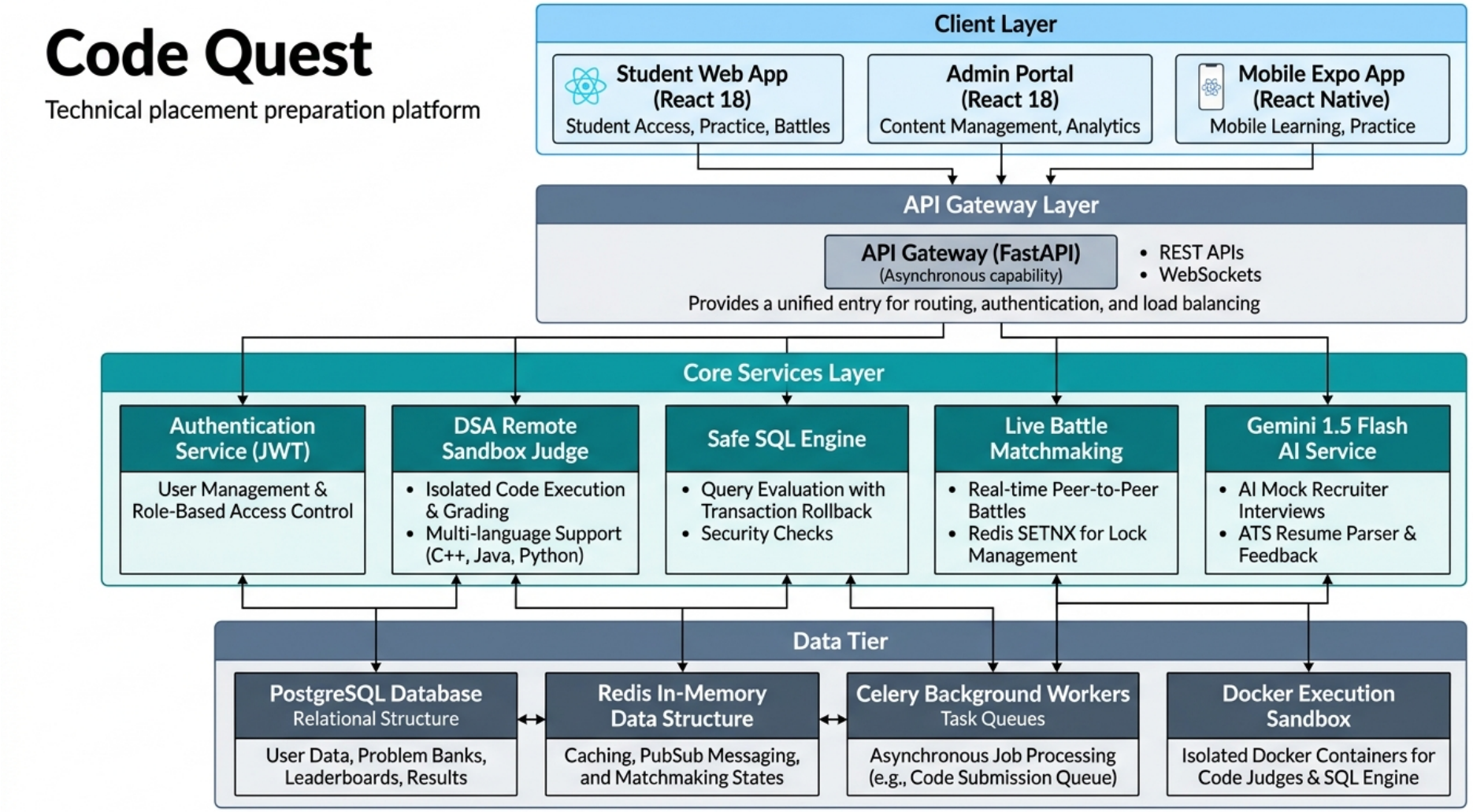


Figure 1: Multi-Tier System Module Architecture (Client Layer, FastAPI Gateway, Core Services, and Distributed Data Tier)

CODE QUEST — UML USE CASE DIAGRAM

Actor Catalog & Functional System Boundaries (Students, Institutional Admins, Sandbox Judge, AI Gemini, Redis)

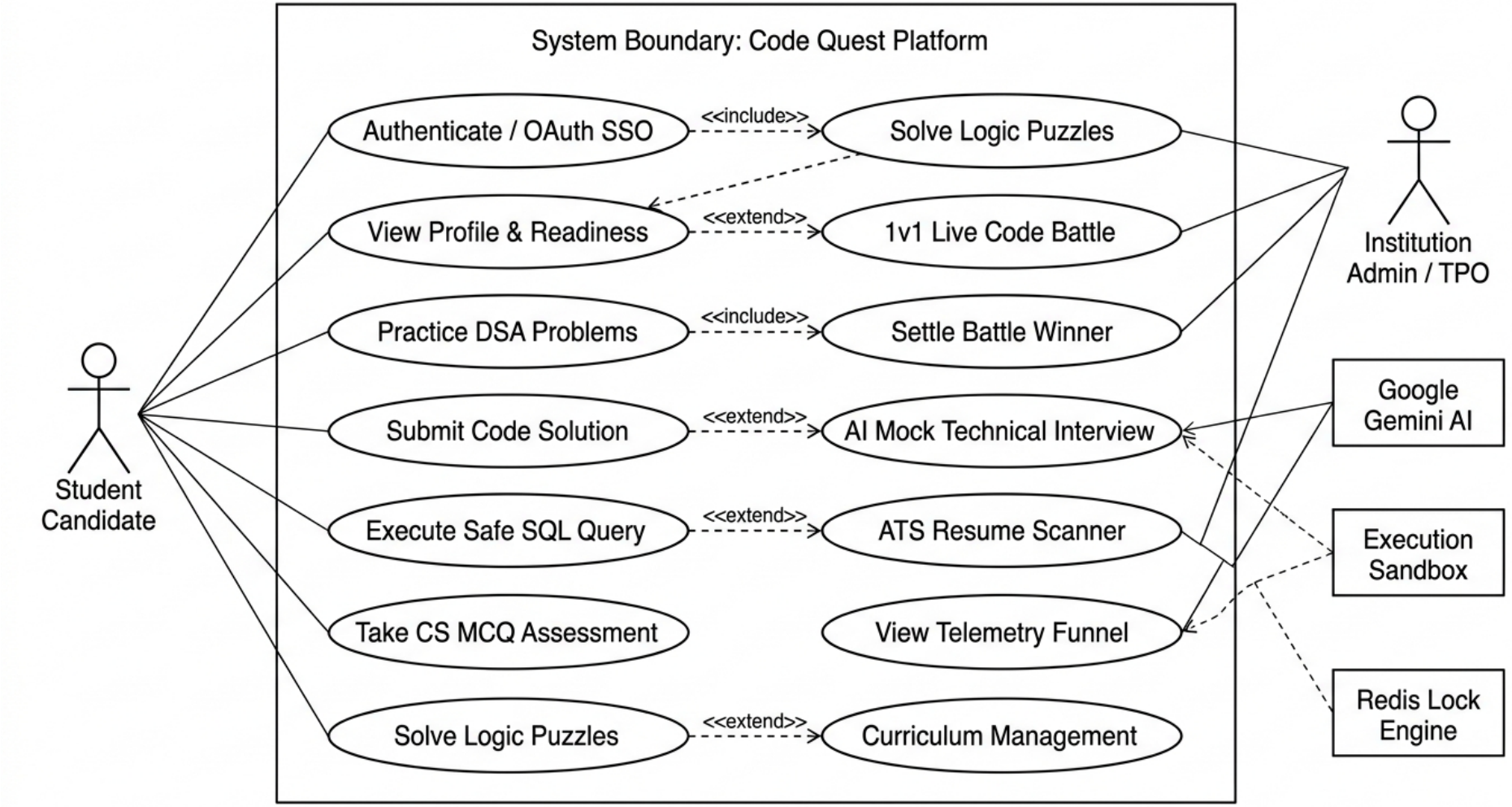


Figure 2: Formal UML Use Case Diagram mapping 16 Functional Use Cases across System Actors

CODE QUEST — DATA FLOW DIAGRAMS (DFD)

Gane & Sarson Multi-Level Decompositions: Level 0 Context Diagram, Level 1 Core Processes, Level 2 Pipelines

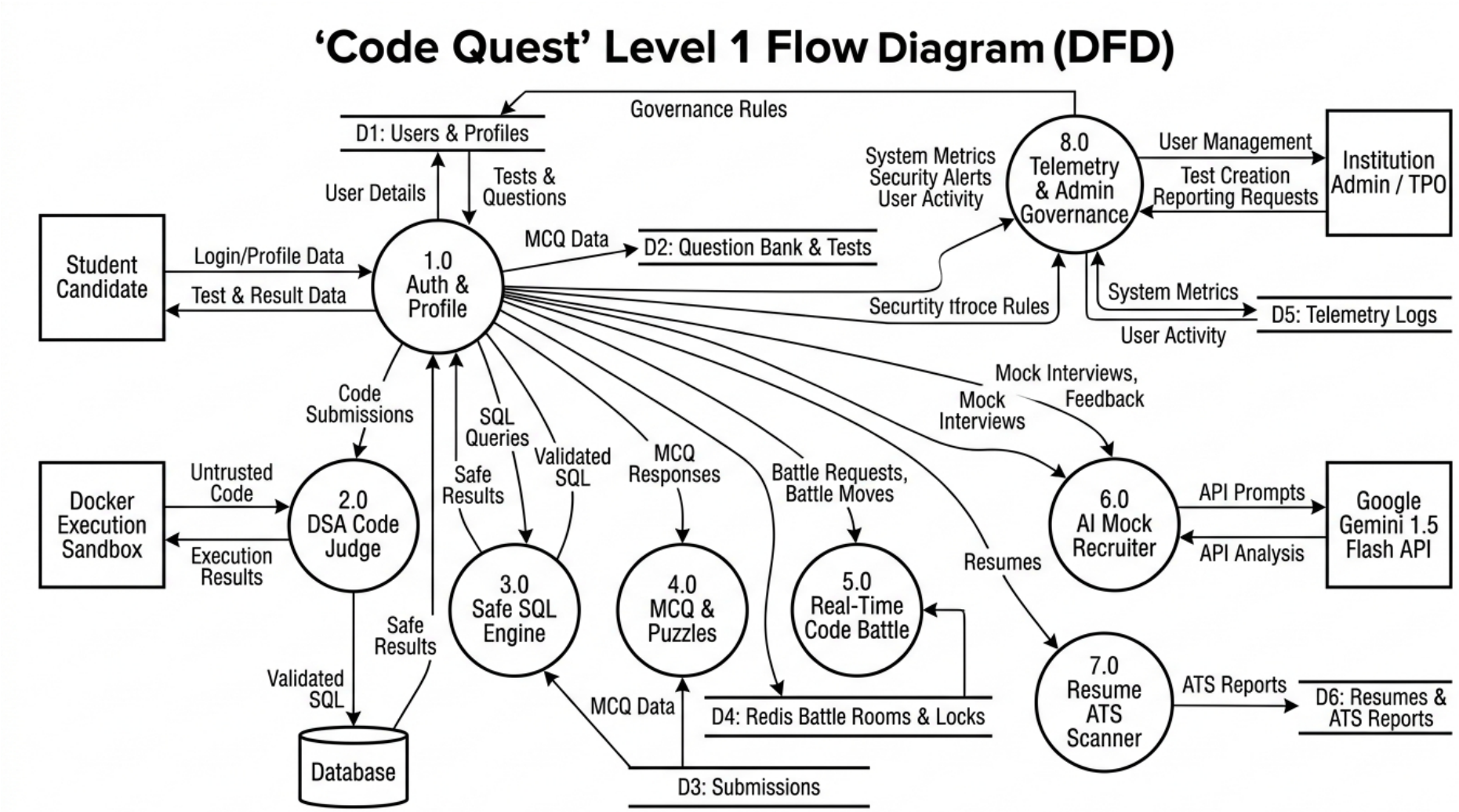


Figure 3: Complete Multi-Level Data Flow Diagram (Level 0 Context, Level 1 Decomposition P1-P8, Level 2 Execution Pipelines)

CODE QUEST — COMPLETE MODULE STATUS GRID

Systematic Implementation, Test Verification, and Production Readiness Audit

#	Module Name	Architectural Responsibility & Features	Technologies & Stacks	Verification / Tests	Status
1	**Auth & RBAC**	Registration, JWT token pair, Google OAuth 2.0 SSO, bcrypt hashing, profile readiness tracking.	FastAPI, PyJWT, Passlib, SQLAlchemy	test_auth.py (5/5 Passing)	**100% COMPLETED**
2	**DSA Code Arena & Judge**	Monaco editor, multi-language sandbox judge (Python, C++, Java, JS), timeout & memory constraints.	Docker Alpine, Subprocess, Cgroups	test_problems_topics.py (4/4)	**100% COMPLETED**
3	**Safe SQL Playground**	Interactive query execution on real schemas, destructive keyword blocking, unconditional transaction ROLLBACK.	PostgreSQL, SQLite, Transaction API	test_mcqs_sql.py (4/4 Passing)	**100% COMPLETED**
4	**CS Domain Assessments**	Timed MCQs covering OS, DBMS, Networks, OOP, and Aptitude with negative marking penalties.	FastAPI, Pydantic, SQLAlchemy	test_mcqs_sql.py (Passing)	**100% COMPLETED**
5	**Algorithmic Puzzles**	Lateral thinking & math challenges, 3-tier progressive hint unlocking with XP point deductions.	FastAPI, React 18, TailwindCSS	test_puzzles_and_battles.py	**100% COMPLETED**
6	**Real-Time Code Battles**	1v1 & multiplayer matchmaking via WebSockets, countdown sync, atomic Redis SETNX distributed lock.	WebSockets, Redis 7+ SETNX	test_live_battles_ws.py	**100% COMPLETED**
7	**AI Mock Technical Recruiter**	Multi-turn conversational mock interview simulation with Google Gemini 1.5 Flash & rubric scorecards.	Google Gemini 1.5 Flash API	ai_features.py (Verified)	**100% COMPLETED**
8	**AI ATS Resume Analyzer**	PDF text extraction, semantic role matching, missing keyword gap analysis, STAR bullet rewriter.	PyPDF, Gemini 1.5 Flash, Celery	test_resume_ats.py (Passing)	**100% COMPLETED**
9	**Admin Telemetry & TPO Portal**	Student readiness indexing, DAU/WAU telemetry, drop-off funnels, curriculum CMS, account moderation.	React 18, TypeScript, Recharts	test_admin_and_dashboard.py	**100% COMPLETED**
10	**Student Web Client**	Responsive SPA web client, Monaco code editor, dynamic XP/streak gamification, theme customization.	React 18, Vite, TailwindCSS, Axios	npm run build (0 errors)	**100% COMPLETED**
11	**Mobile Application**	Cross-platform mobile client for on-the-go practice, MCQs, and roadmap progression.	React Native, Expo SDK, TypeScript	codequest.apk built	**100% COMPLETED**